

All models are evaluated and compared using important classification metrics such as **Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score**. These metrics help in identifying the model that provides the most accurate and reliable predictions for the dataset.

After evaluating the performance of all trained models, the best-performing model is selected based on its overall classification results and generalization capability. The finalized model is then serialized using **pickle.dump()** and stored as **model.pkl** so that it can be reused without retraining during deployment in the Flask web application.